

7 (a) (i) e.g. electron / particle diffraction B1 [1]

(ii) e.g. photoelectric effect B1 [1]

(b) (i) 6 A1 [1]

(ii) change in energy = $4.57 \times 10^{-19} \text{ J}$

$$\lambda = hc / E$$

C1

$$= (6.63 \times 10^{-34} \times 3.0 \times 10^8) / (4.57 \times 10^{-19})$$

$$= 4.4 \times 10^{-7} \text{ m}$$

A1 [2]

8 (a) splitting of a heavy nucleus (not atom/nuclide)

M1